



Temperature as a central driver of *Aedes koreicus* biology, ecology and vector potential

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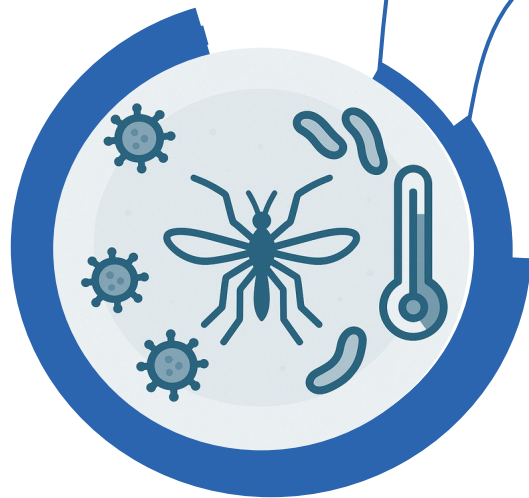


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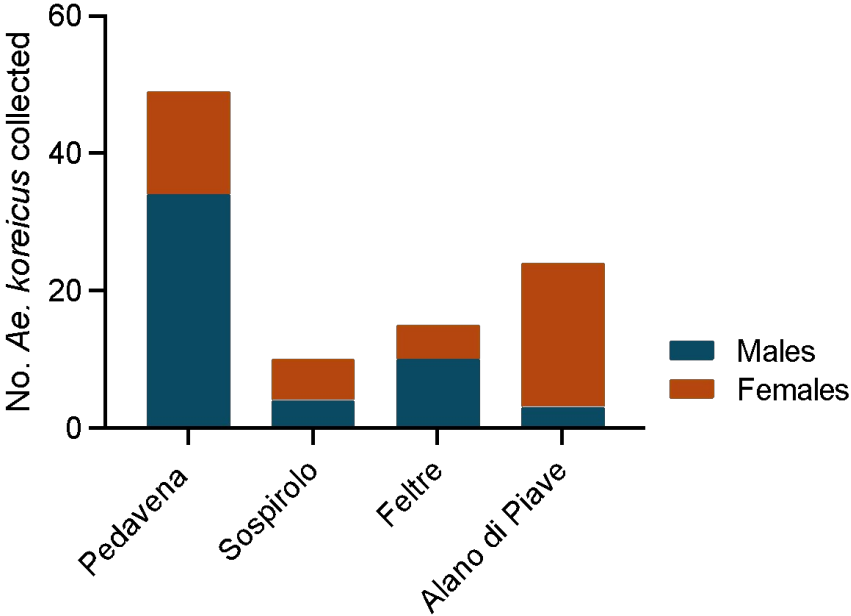


Thermal biology of *Ae. koreicus*

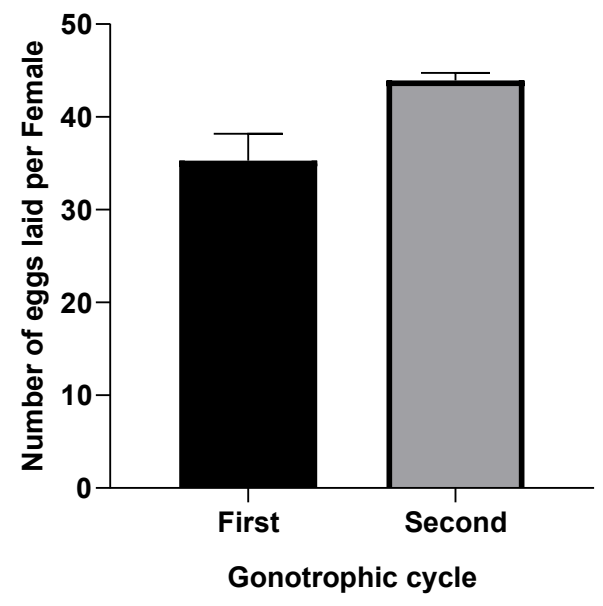
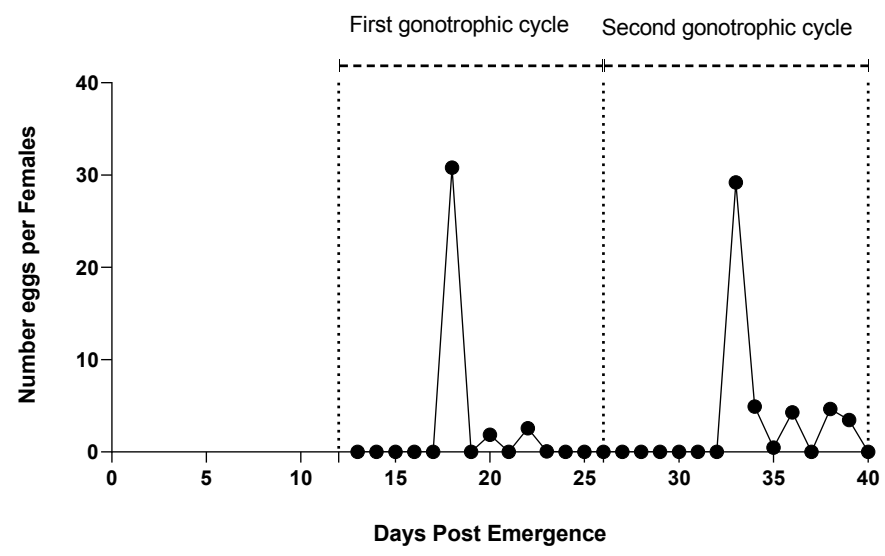
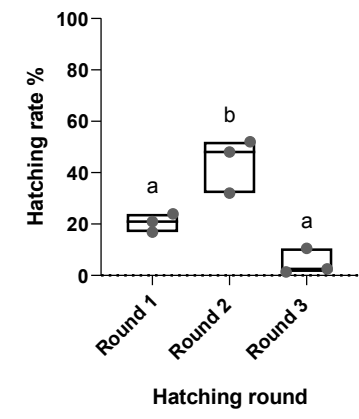
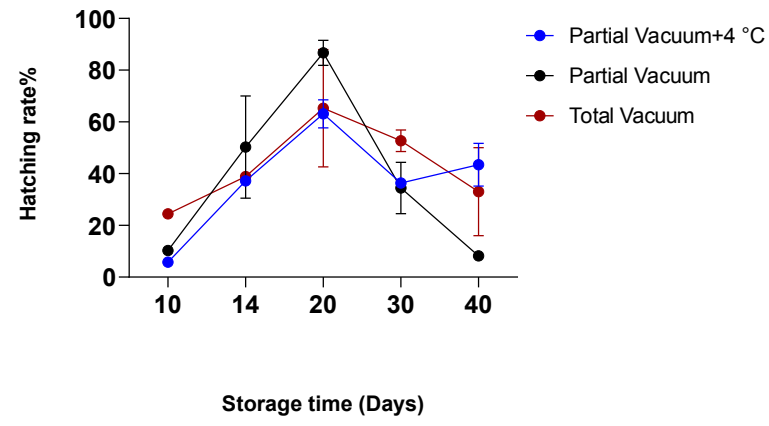
Temperature-dependent vector competence
of *Ae. koreicus*

Interaction of *Asaia* (the main symbiont of *Ae. koreicus*)
with its host

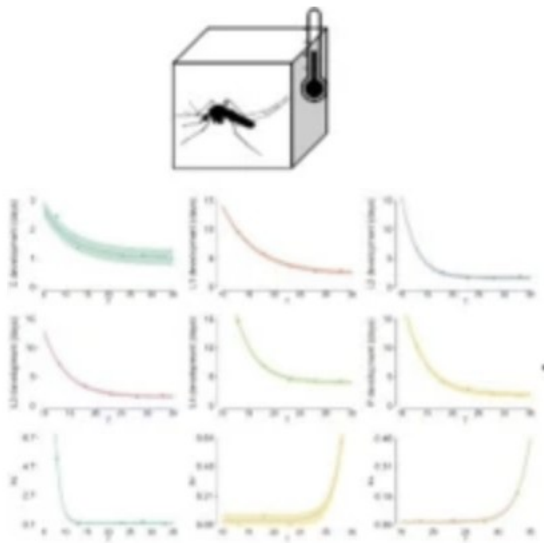
Collections and colony establishment : an interesting task



General Life Traits

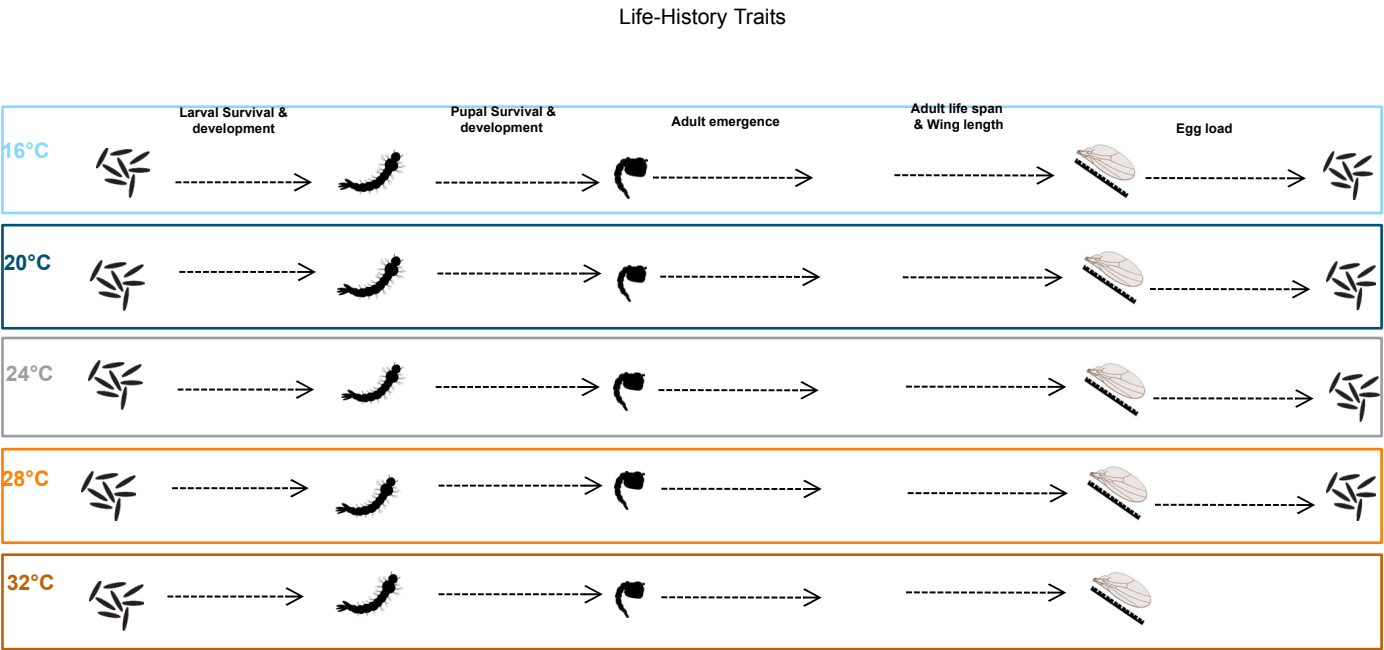


If *Ae. koreicus* has a lower optimum can this cause geographic expansion and the likelihood of prolonged mosquito seasons in invaded regions?

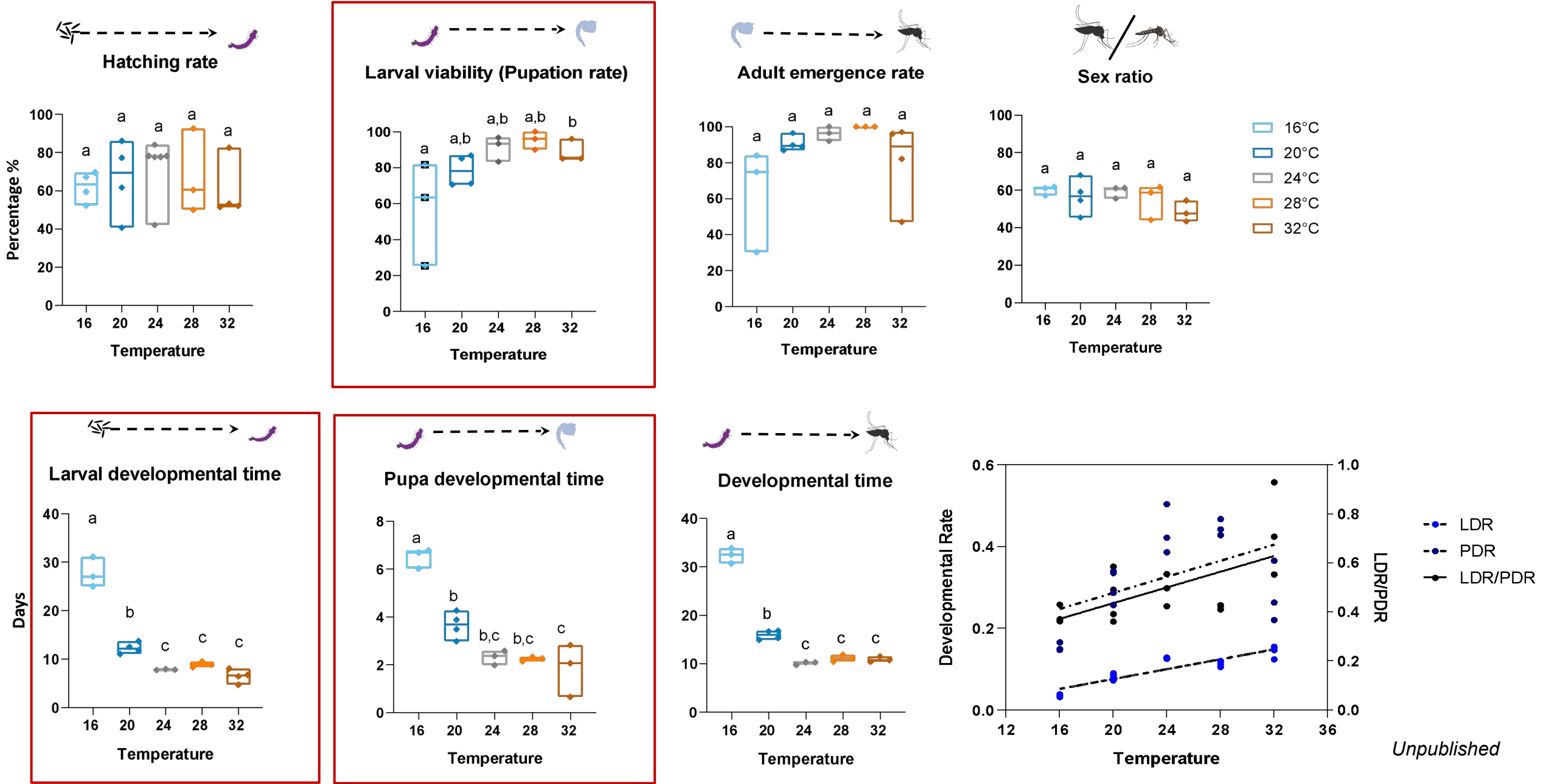


Marini et al., 2019

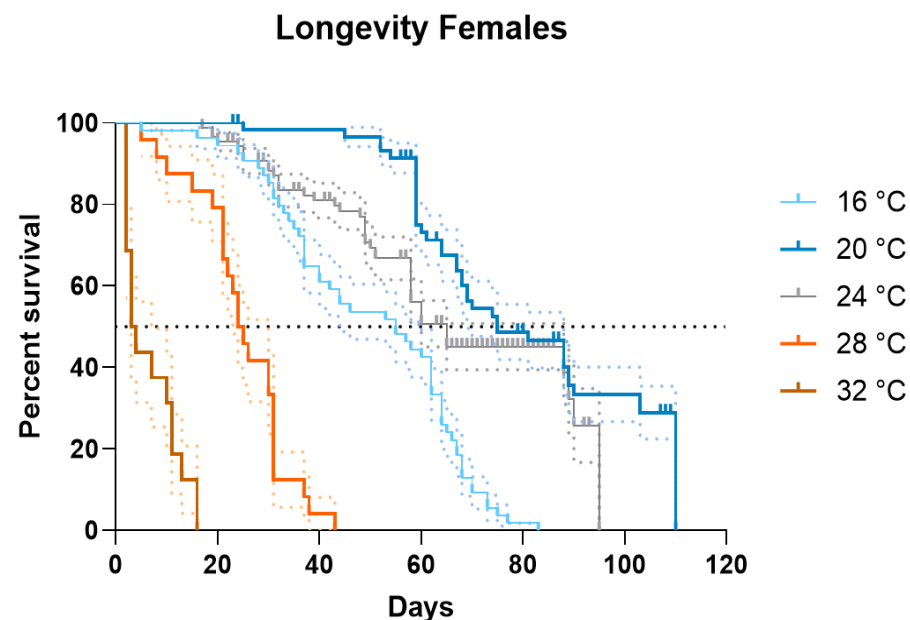
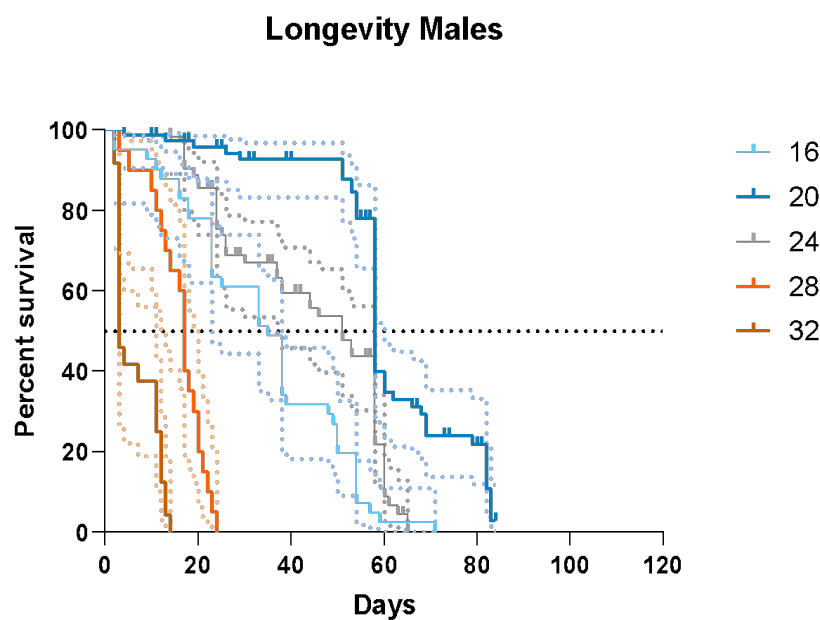
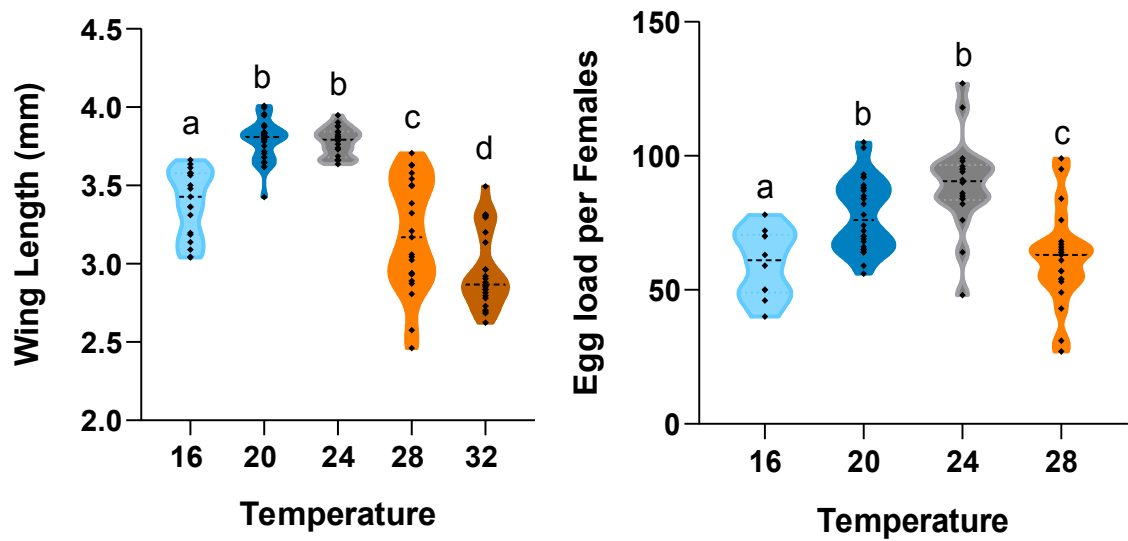
Favourable range: 23–28 °C



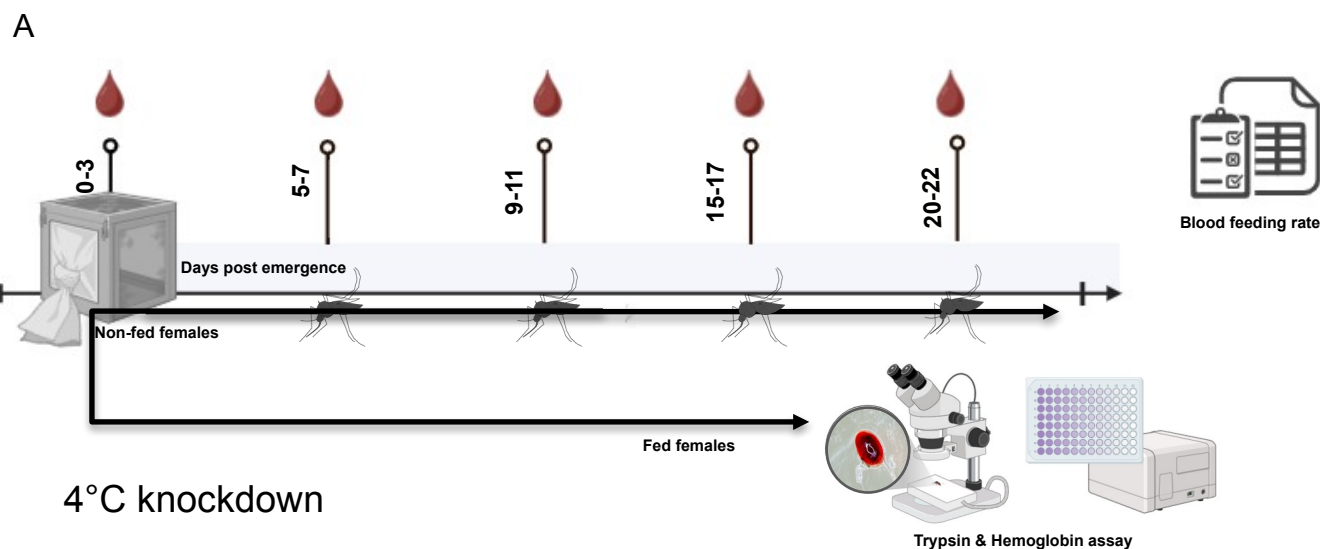
Juvenile stages are more resilient to higher temperature



Mature stages are less resilient to higher temperature

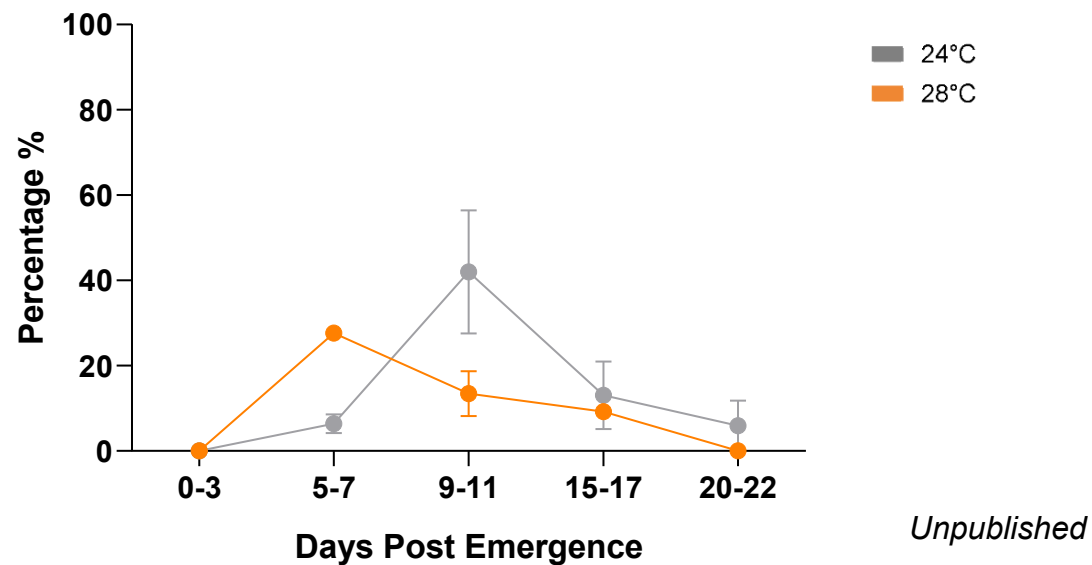
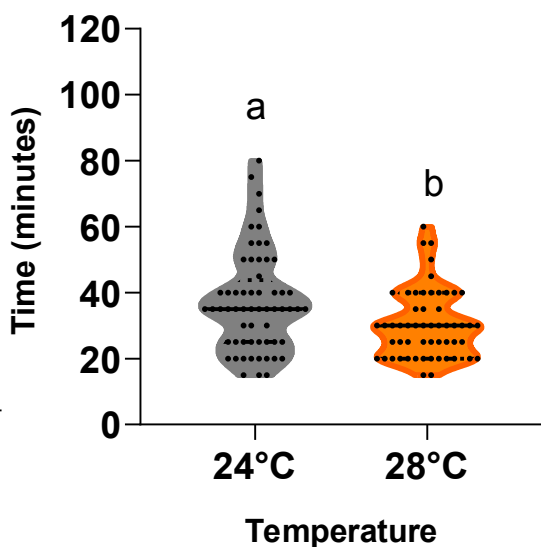
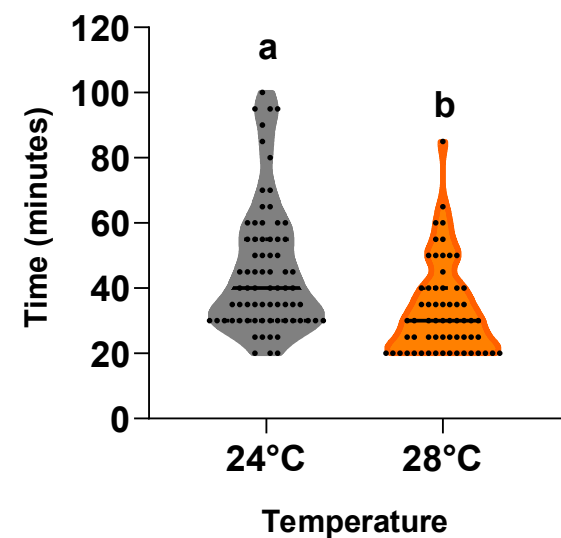


Temperature affects blood feeding behaviour and shifts the blood feeding peak



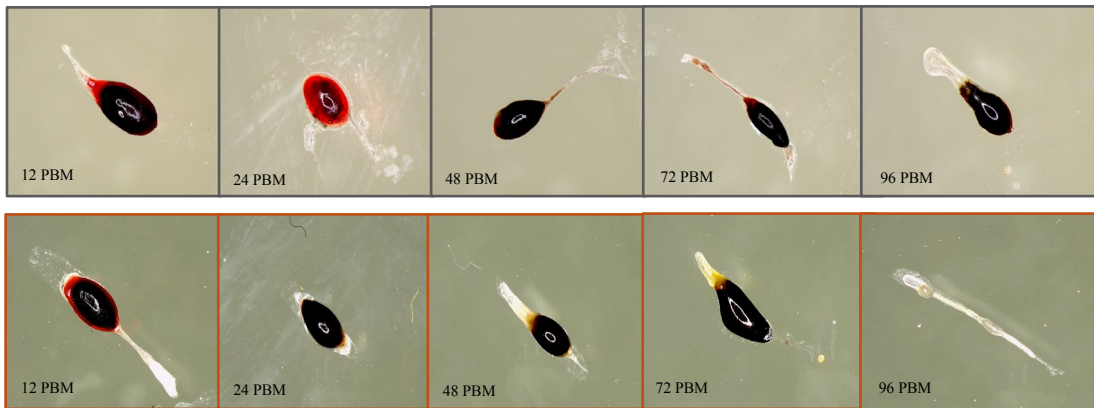
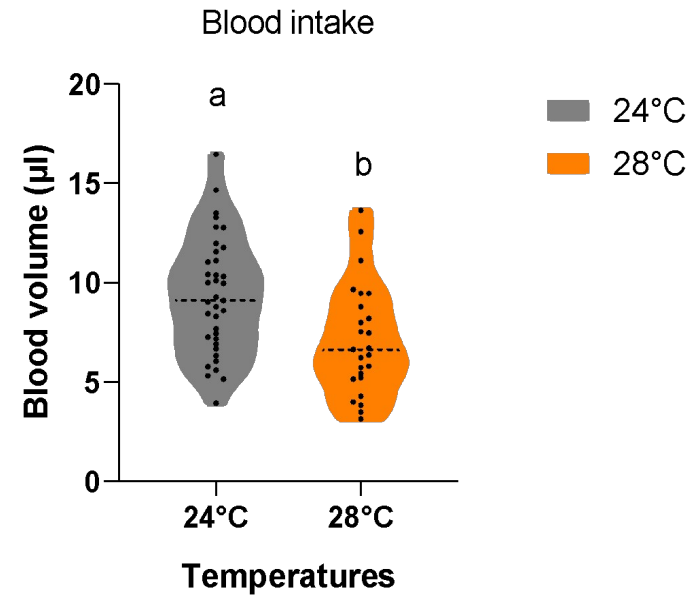
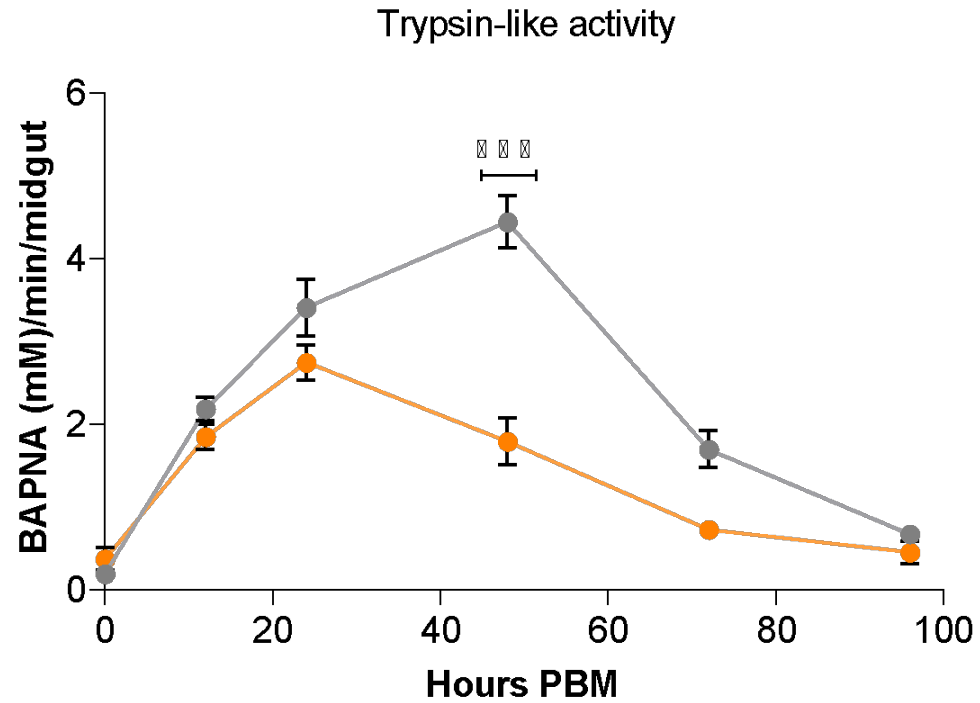
Males

Females



Unpublished

Temperature affects the blood digestion pattern in *Ae. koreicus*



Could blood feeding metabolism affect the dynamics of viral entry and dissemination in mosquito hosts?

Temperature affects dormancy patterns of *Ae. koreicus*

Saving accounts for Life

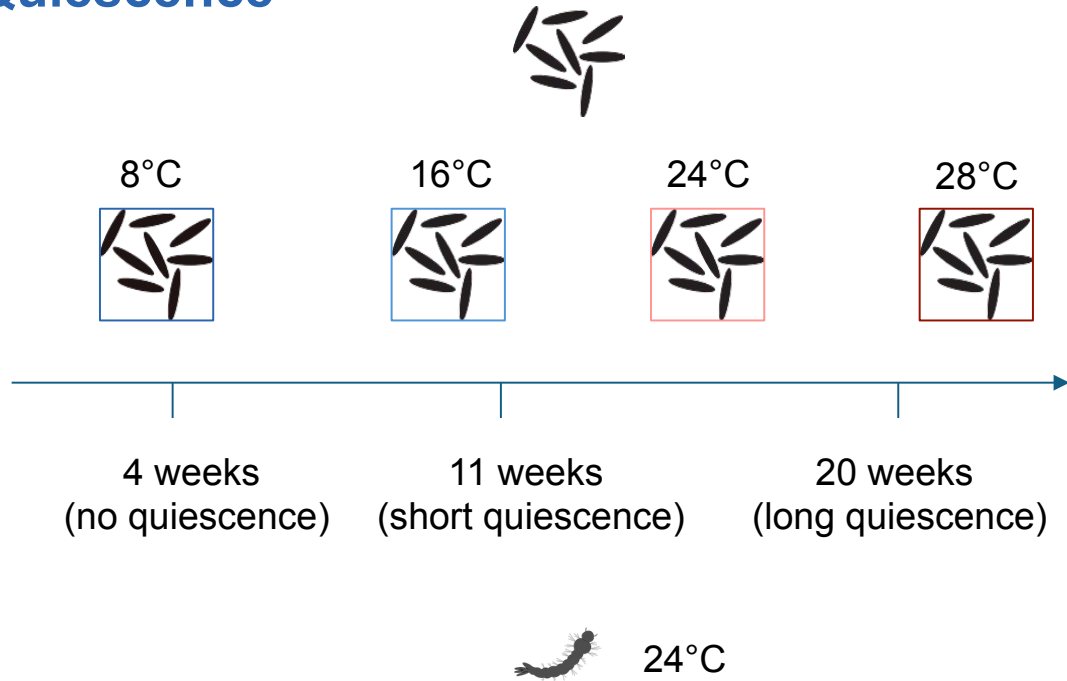
Quiescence = reactive dormancy

Immediate, reversible response to **unfavorable conditions**

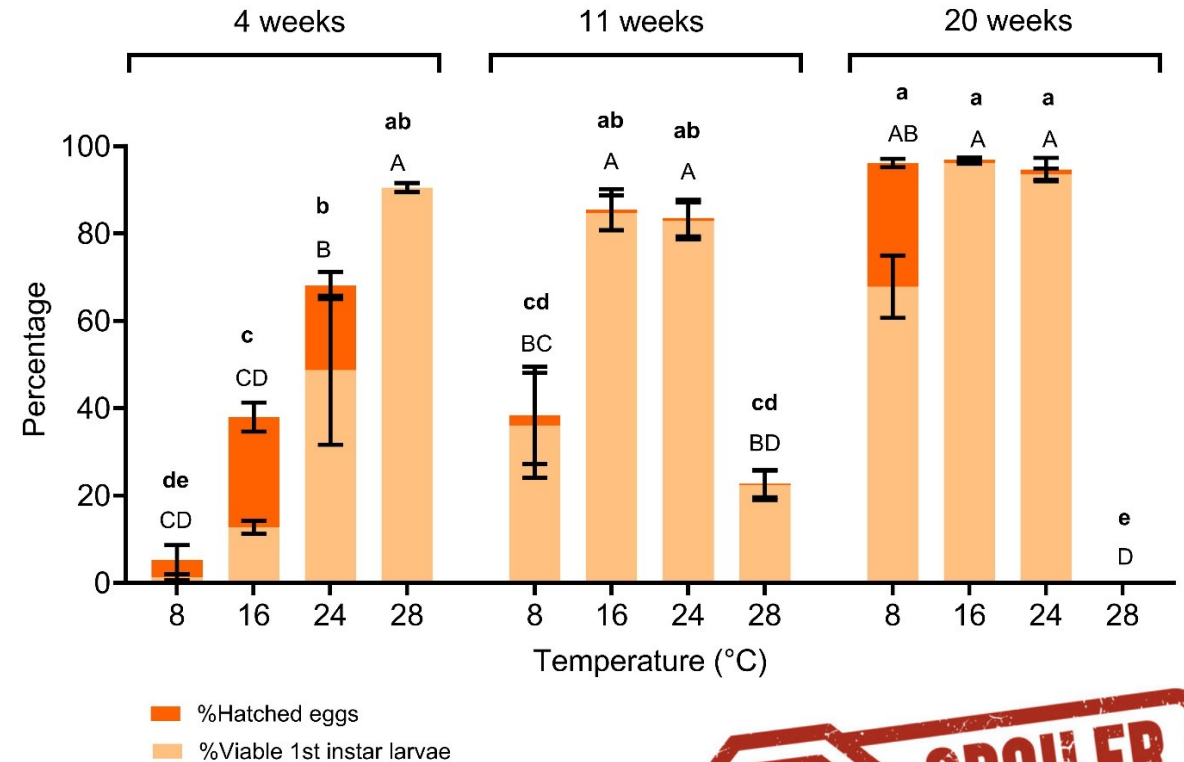
Diapause = anticipatory dormancy

Programmed developmental arrest initiated **before** adverse conditions

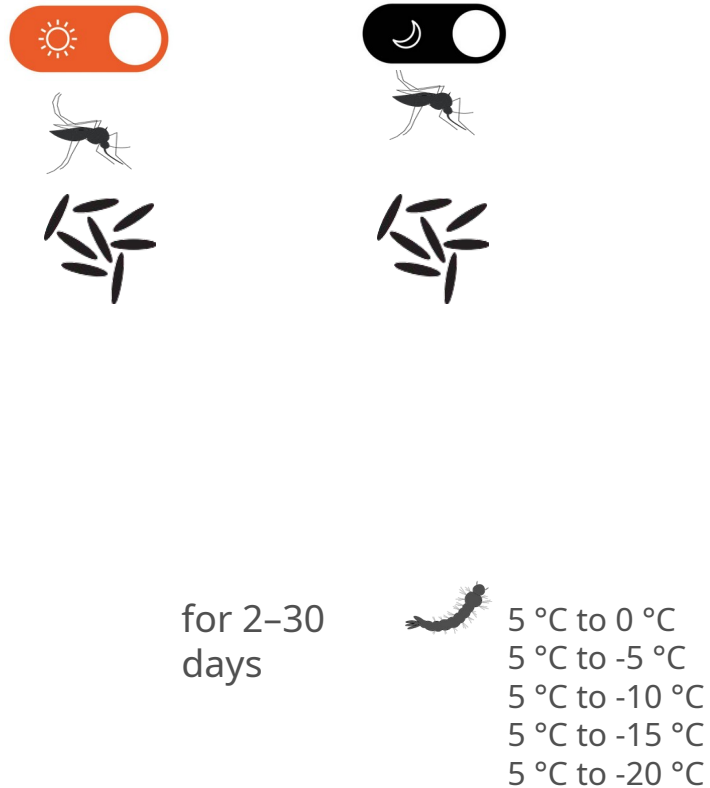
Quiescence



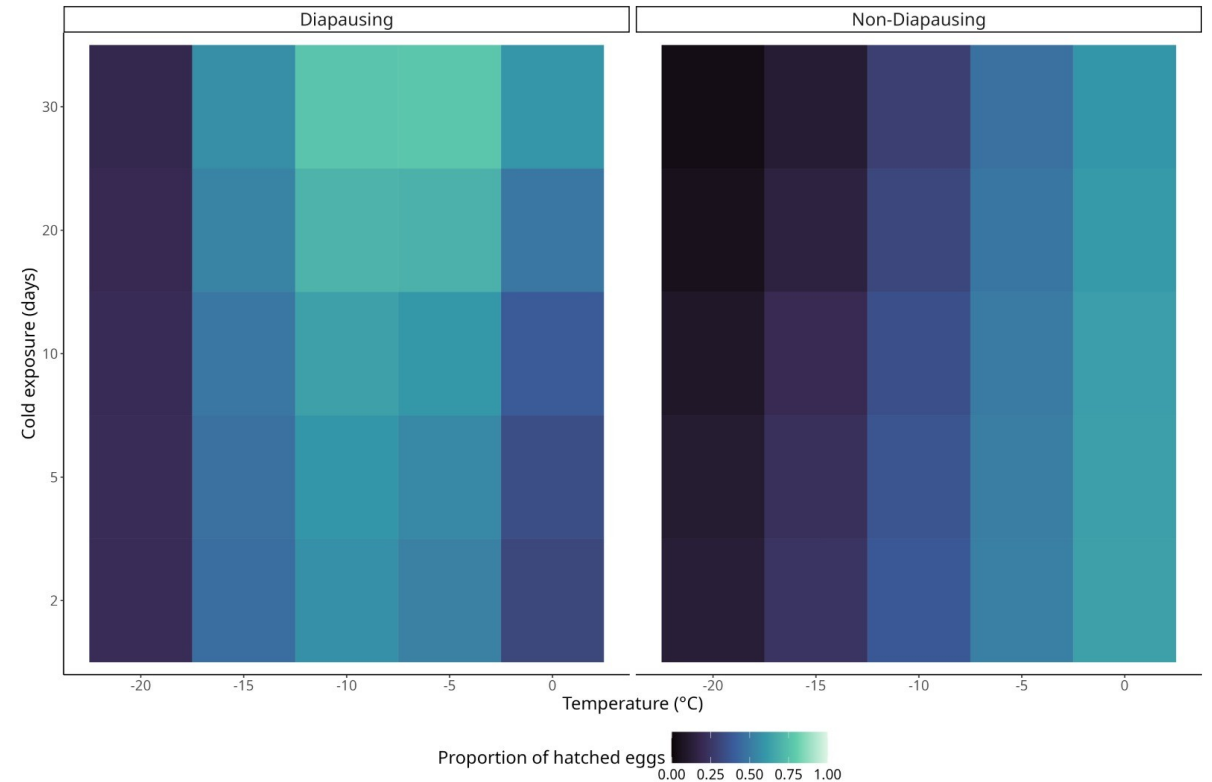
- Juvenile stages development
- Adult life history traits
- Adult energy reserves
- Microbiota perturbation

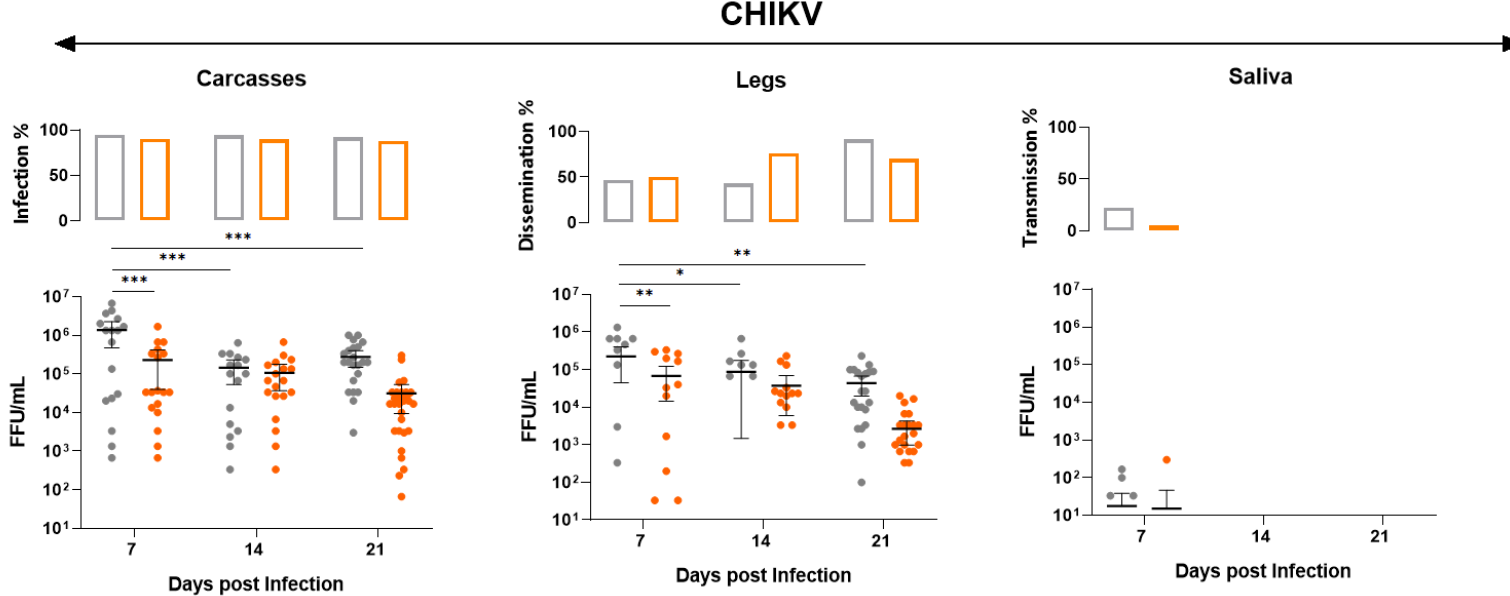


Diapause



- Cold exposure reduces eggs hatching especially in non-diapausing eggs.
- Longer cold exposure increase hatching success in diapausing eggs.





Higher T^* does not enhance CHIKV transmission in our system.

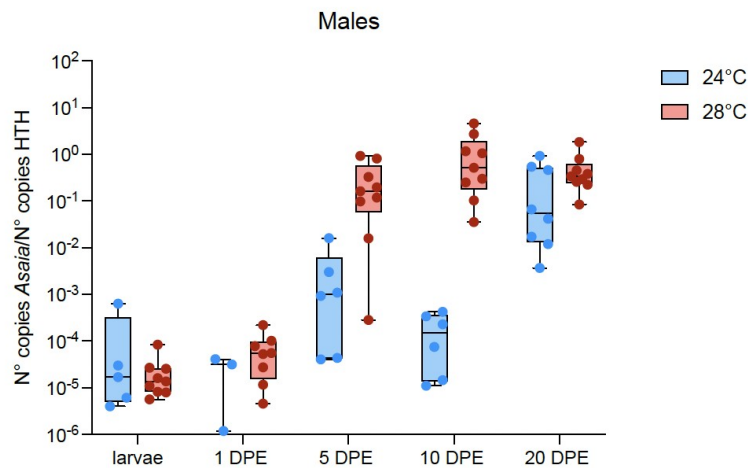
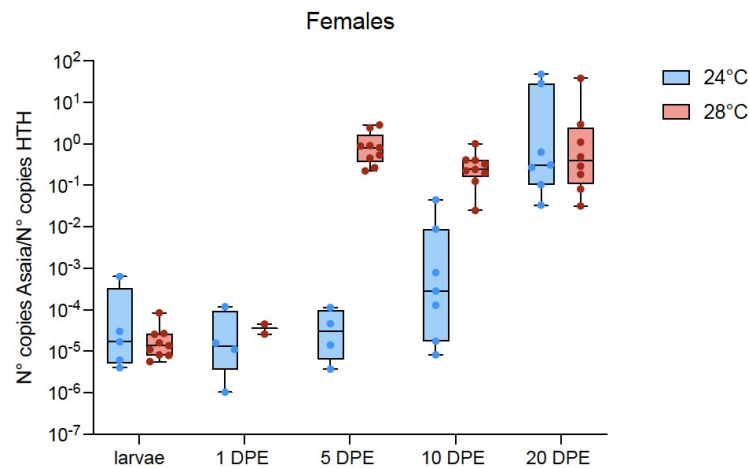
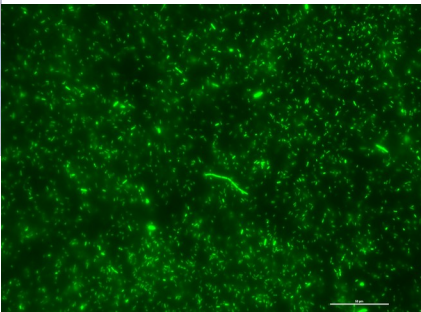
Non-linear temperature effects on vector competence

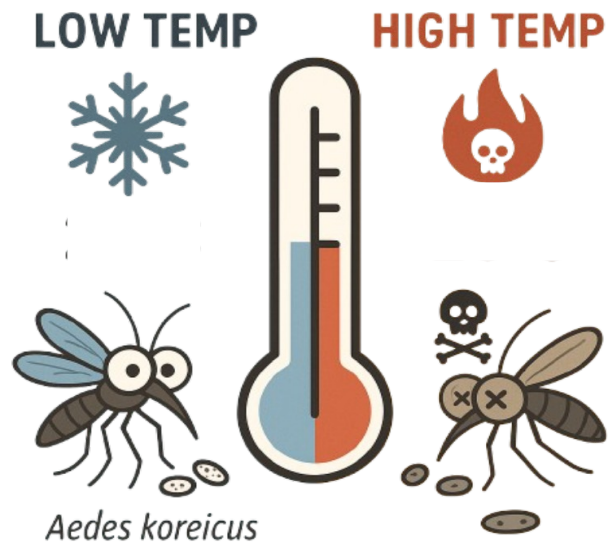
Temperature affects the density and persistance of native symbionts

Asaia bacteria



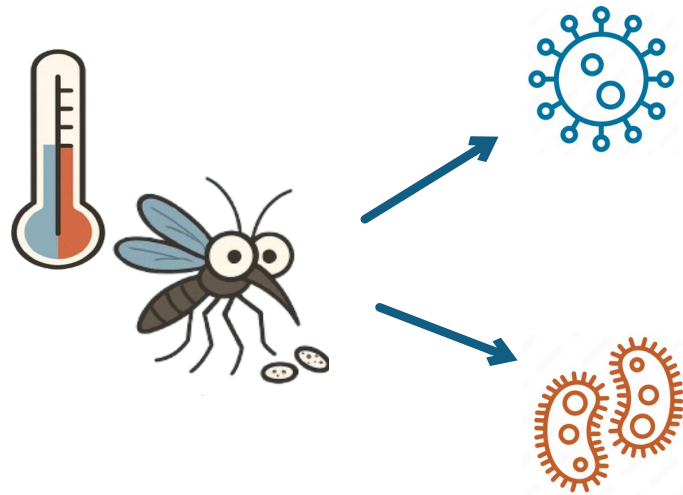
GFP-expressing *Asaia* isolated from *Ae. koreicus*





Temperature affects *Ae. koreicus* **more during the adult stage than the larval stage.**

The **optimal** temperature for *Ae. koreicus* performance is around **20 - 24 °C.**



Temperature has an effect (direct or indirect) on **vector competence and stability of native host-bacteria interactions.**



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Thanks for the attention



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